Community Standard Justification: 26-XXX

TITLE: 3D Tiles 2.0 Community Standard Justification

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1. Introduction

This document presents the justification for submission of the next revision to the OGC 3D Tiles Community Standard for consideration by the OGC Technical Committee (TC). This justification forms the basis for TC review and vote to approve initiation of a Work Item to update the OGC 3D Tiles Community Standard to version 2.0.

The submitters agree to abide by the TC Policies and Procedures and OGC Intellectual Property Rights Policy (https://www.ogc.org/about/policies/) during the processing of this submission.

Once approved, the Community standard Work Item defined by this document is valid for six (6) months.

2. Overview of proposed submission

3D Tiles 1.0 is an OGC Community Standard for sharing, visualizing, fusing, and interacting with massive heterogenous 3D geospatial content across desktop, web, and mobile applications.

3D Tiles 1.1 evolved 3D Tiles to enhance the handling of semantic metadata, enable implicit tiling, and introduced direct references to glTF files allowing for better integration with the glTF ecosystem.

3D Tiles 2.0 is a proposed revision to the 3D Tiles community standard designed for streaming high-resolution semantically rich 3D geospatial data. 3D Tiles 2.0 continues the trend started in 3D Tiles 1.1 and further aligns 3D Tiles with the glTF ecosystem by baselining upon glTF 2.1's new external asset referencing for spatial subdivision. Several deprecated 3D Tiles 1.0 features are being removed, such as legacy 3D Tiles 1.0 file formats.

The primary proposed scope for OGC 3D Tiles 2.0 includes:

- Baselining on glTF 2.1

The secondary proposed scope for OGC 3D Tiles 2.0 includes:

- Architectural, Engineering, and Construction (AEC) Rendering Extensions
- Time-Dynamic 3D Tiles
- Voxels
- Vector Tiles
- 3D Gaussian Splatting

A tileset that conforms to 3D Tiles 1.1 will also conform to 3D Tiles 2.0, except for the asset.version field. 3D Tiles 2.0 implementations will accept tilesets declaring asset.version of either 1.1 or 2.0. 3D Tiles 2.0 tilesets should declare a glTF 2.1 document as the entry document but may use tileset JSON as the entry document with the 3D Tiles 1.1 feature set excluding removed features.

3D Tiles 2.0 is expected to be submitted for OGC ratification in Q4 2026. Reference implementers are committed to updating their implementations with the final release of the specification. OGC member review of the specification after approval of the Justification document will be valuable in finalization.

Background

3D Tiles, announced at SIGGRAPH in 2015 and published as an OGC community standard in 2019, has served as a foundation for sharing, visualizing, and interacting with massive 3D geospatial content. Its flexible ecosystem has enabled the development of apps, exporters, APIs, and engines, supporting a broad range of geospatial use cases.

With growing availability of semantically rich 3D geospatial data and increased interest in digital twins and 3D mapping, 3D Tiles has continued to evolve to meet new needs. Version 1.1 of 3D Tiles is the key data standard supporting the U.S. Army's One World Terrain (OWT) program, used to store and deliver geospatial terrain data to the U.S. Army's Synthetic Training Environment (STE) Point of Need (PoN) training systems.

3D Tiles 2.0 further advances the standard by building on glTF 2.1's advanced features, such as complex scenes and spatial subdivision, incorporating enhancements to Architecture, Engineering, and Construction (AEC) rendering, and support for Voxels, Vector Tiles, and 3D Gaussian Splatting. These additions support growing workflows and domains within the 3D Tiles community, including AEC, Aerospace, Defense, Entertainment, and e-Commerce.

2.1 Baselining on glTF 2.1

From the beginning, 3D Tiles has co-evolved with glTF. 3D Tiles 2.0 takes that a step further and is now baselined *on* glTF, bringing the latest graphics capabilities, and its entire supporting ecosystem, into the standard.

3D Tiles 2.0 uses glTF 2.1, together with a set of geospatial extensions, as the foundation for spatial subdivision. Tilesets express scene hierarchy, external asset referencing, and bounding volume hierarchies using the glTF 2.1 external assets model. Implementations provide hierarchical level-of-detail and support the geospatial glTF extensions identified in the Normative References section.

3D Tiles 2.0 retains responsibility for the core geospatial concepts that define the standard, including tiling schemes (such as S2 and other discrete global grid systems), region bounding volumes, and coordinate reference system and ellipsoid declarations. New glTF extensions that operate across the scene graph are supported by 3D Tiles 2.0 clients without requiring additional 3D Tiles–specific handling.

A glTF 2.1 file with two external references, rendered in CesiumJS.

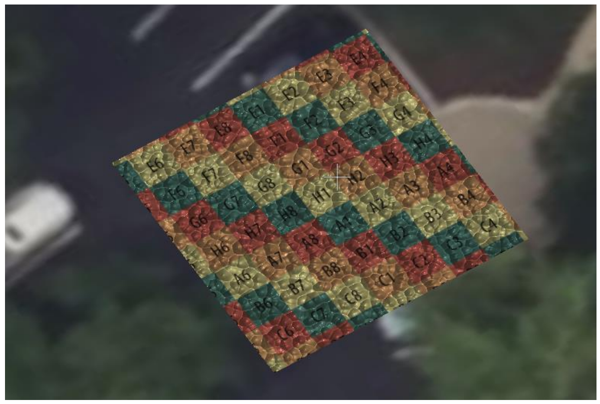
2.2 Architectural, Engineering, and Construction Rendering Extensions

3D Tiles 2.0 introduces a dedicated suite of AEC rendering extensions: edge and wireframe visibility for crisp component boundaries, configurable line and point styles for the full richness of CAD annotation, z-fighting mitigation for co-located surfaces like roads and labels, and consistent texture LOD so that hatching and patterns stay legible at every zoom level.

3D Tiles 2.0 supports edge rendering for AEC workflows through the glTF extensions *KHR_mesh_primitive_restart* and *EXT_mesh_primitive_edge_visibility*. Implementations recognize both extensions in tile content and render the content according to the extension definitions or report the extension as unsupported, in accordance with glTF 2.1.

For styling points and lines in CAD content, 3D Tiles 2.0 supports the *BENTLEY_materials_line_style* and *BENTLEY_materials_point_style* glTF extensions. These extensions define the properties used to represent hidden and conceptual elements common in civil and engineering rendering workflows.

Implementations support EXT_textureInfo_constant_lod to maintain consistent texture pattern scale across zoom levels, and may support *BENTLEY_materials_planar_fill* to mitigate z-fighting between co-located surfaces such as labels, roads, and overlapping infrastructure.



Sample plane using EXT_textureInfo_constant_lod rendered in iTwin.js.

2.3 Time-Dynamic 3D Tiles

3D Tiles 2.0 supports time-dynamic tiling, enabling multiple representations of the same spatial region within a single tileset. This is useful for temporal states of terrain, design-model revisions, or alternative geometry representations such as point clouds alongside meshes.

Any tile may carry one or more attribute key/value tag pairs. Not every tile needs to be tagged; untagged tiles are treated as unconditional. Clients select tile content by evaluating these tags against a selection expression supplied at runtime. Where no selection expression is provided, clients render the default content declared by the tileset author.

Authoring tools can use time-dynamic 3D Tiles to publish incremental updates without rewriting an entire tileset.

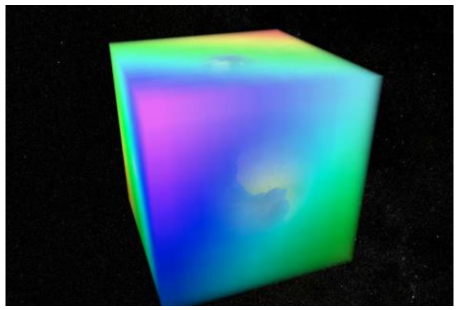
3D Tiles of construction site data highlighting changes over time using Time-dynamic 3D tiles in CesiumJS.

2.4 Voxels

3D Tiles 2.0 brings volumetric data into the standard, supporting both dense and sparse volumes, multiple compression strategies, and seamless integration with the same implicit tiling mechanism that already powers efficient terrain streaming.

Voxel tilesets use the *EXT_primitive_voxels* together with its dependencies — *3DTILES_shape_ellipsoid_region*, *3DTILES_shape_cylinder_region* glTF 2.1 extensions in combination with the implicit tiling mechanism defined in 3D Tiles 2.0.

Both dense and sparse voxel volumes are supported. Voxel payloads may be uncompressed or compressed using gzip, the glTF *EXT_meshopt_compression* extension, the glTF *KHR_meshopt_compression* extension, or a combination of these methods.

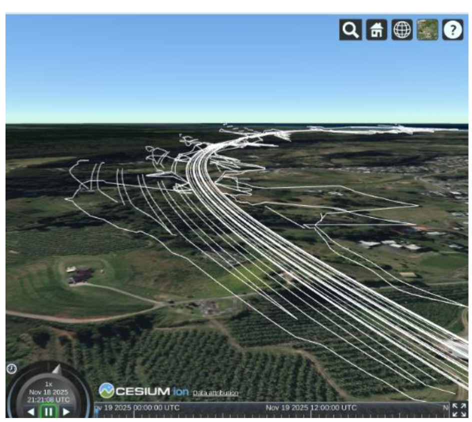


Procedurally generated box voxels in CesiumJS.

2.5 Vector Tiles

3D Tiles 2.0 supports both 2D and 3D vector data using a 3D-native data model consistent with the rest of the standard and with glTF 2.1. Relevant dependencies include *3DTILES_content_gltf_vector* for 3D Tiles, and *KHR_mesh_primitive_restart* and *EXT_mesh_polygon* for glTF. Tilesets can represent road centerlines, building footprints, large-scale map features, and AEC design linework as vector data. Payloads may be uncompressed or compressed using glTF *EXT_meshopt_compression* or *KHR_meshopt_compression*.

3D Tiles 2.0 is designed for efficient transmission and rendering of 3D vector data, which differentiates it from established vector formats like Mapbox Vector Tiles and GeoJSON. Implementations may provide interoperability paths to and from those formats where appropriate. Styling of vector content uses the same declarative styling language defined for other 3D Tiles content types.



Proposed 3D highway-design linework as vector tiles.

2.6 3D Gaussian Splatting

3D Tiles 2.0 supports 3D Gaussian Splatting content via the *KHR_gaussian_splatting* glTF extension. Implementations that support the Gaussian Splatting conformance class recognize this extension. Implementations may also support *KHR_gaussian_splatting_compression_spz_2*, which defines compression of splat data using the SPZ format; uncompressed splat content may be used where compression is not required.

Splat tilesets are compatible with the hierarchical level-of-detail mechanism of 3D Tiles, so clients can stream and cull splat content using the same traversal rules as other tile content types.

Photorealistic 3D Gaussian splats of Microsoft's campus rendered in CesiumJS with 3D Tiles 2.0

2.7 Normative References

The following documents are referred to in the text in such a way that some or all their content constitutes the requirements of this document. For dated references, only the edition cited applies; for undated references, the latest edition applies.

- Khronos Group, glTF 2.1 Specification
- Bentley, 3DTILES_tileset glTF extension
- OGC 18-010r7, Geographic information — Well-known text representation of coordinate reference systems
- ISO/IEC Directives, Part 2 (by way of OGC Policy Directive 49)"

2.8 Deprecated Features

The following features are deprecated in 3D Tiles 2.0:

- Tileset JSON is deprecated in favor of glTF 2.1.

2.9 Removed Features

The following features previously deprecated in 3D Tiles 1.1 are removed in 3D Tiles 2.0:

- Legacy Tile Formats
- Batched 3D Model (b3dm)
- Instanced 3D Model (i3dm)
- Point Cloud (pnts)
- Composite (cmpt)
- Batch Table
- Feature Table
- The tileset.properties dictionary.

3. Relationship to other OGC standards

3D Tiles normatively references:

- OGC 18-010r7, Geographic information — Well-known text representation of coordinate reference systems.

4. Alignment with OGC Standards Baseline

Examples of 3D Tiles interoperability within the OGC ecosystem:

OGC API - 3D GeoVolumes provides an API for organizing access to a variety of 2D / 3D content and informally references 3D Tiles as a possible content type. This capability was demonstrated in the OGC 3D Data Container and Tiles API Pilot. The GeoVolumes SWG is exploring tile coordinate-based content retrieval and harmonization with 3D Tiles 1.1 implicit tiling.

vis.gl includes an open-source command-line utility that converts between OGC I3S 1.3 Community Standard and OGC 3D Tiles 1.1 Community Standard.

Github link: https://github.com/visgl/loaders.gl/tree/master/modules/tile-converter

Like 3D Tiles 1.1, 3D Tiles 2.0 can be used as an efficient format for streaming massive city datasets originally stored in CityGML.

CDB 2.0 is exploring using glTF as an alternative model format to OpenFlight. New capabilities in glTF 2.1 like external references and level of detail would be valuable in CDB's option of glTF.

Alignment between 3D Tiles and I3S was explored at the May 2026 OGC Code Sprint, particularly the new capabilities in glTF 2.1.

Refer to the original OGC 3D Tiles Community Standard Work Item Justification document OGC 16-123 for additional examples of 3D Tiles interoperability within the OGC ecosystem.

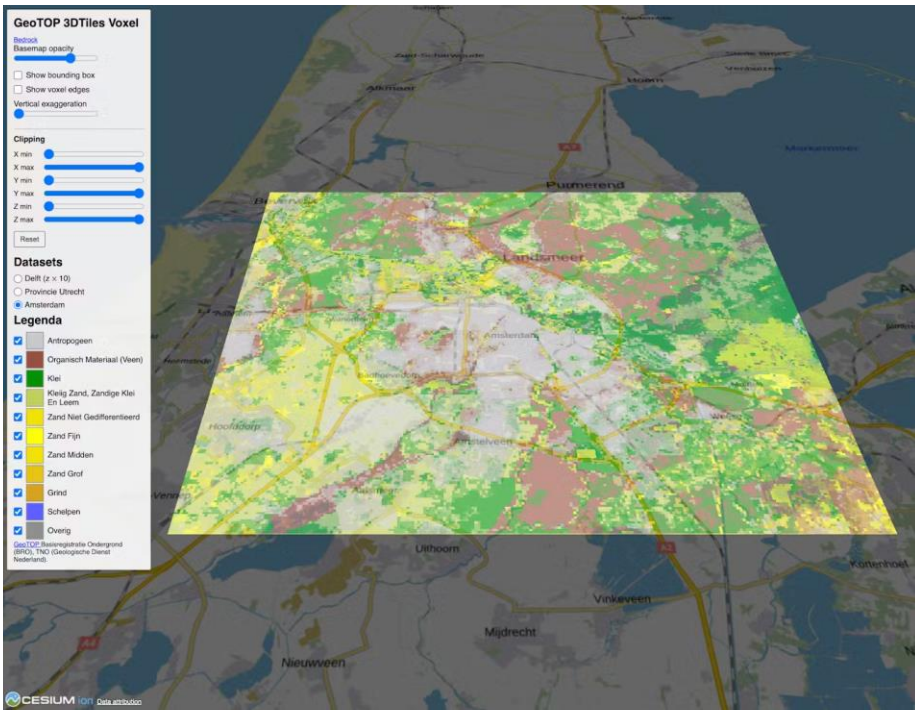
5. Evidence of implementation

The following implementations use the proposed Community standard.

1. Implementation name: Bedrock.engineer

Date of most recent version: Ongoing as the 3D Tiles 2.0 spec finalizes

Implementation description: Uses voxels in 3D Tiles 2.0 to stream large subsurface datasets, to enable geological voxel models to be explored in spatial context directly in the browser.



Bedrock.engineer visualizes GeoTOP's 50m deep subsurface data covering the entire Netherlands in CesiumJS using voxels.

Implementation URL:

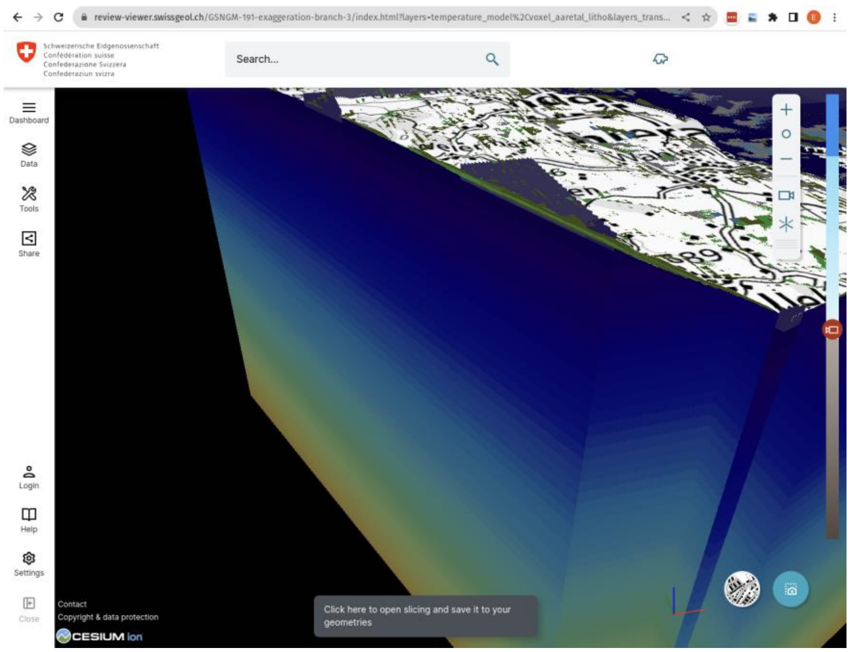
- https://bedrock.engineer/
- https://cesium.com/blog/2026/04/08/bedrock-engineer-streams-subsurface-with-voxels/

Is implementation complete? ☒ Yes ☐ No

2. Implementation name: SwissTopo

Date of most recent version: Ongoing as the 3D Tiles 2.0 spec finalizes

Implementation description: SwissTopo provides several voxel datasets like swissBEDROCK, a comprehensive dataset that provides detailed information on Switzerland's bedrock surface. These datasets are stored and streamed in 3D Tiles 2.0 using voxels.



Subsurface ground composition classification in Switzerland represented using 3D Tiles 2.0 Voxels and CesiumJS.

Implementation URL:

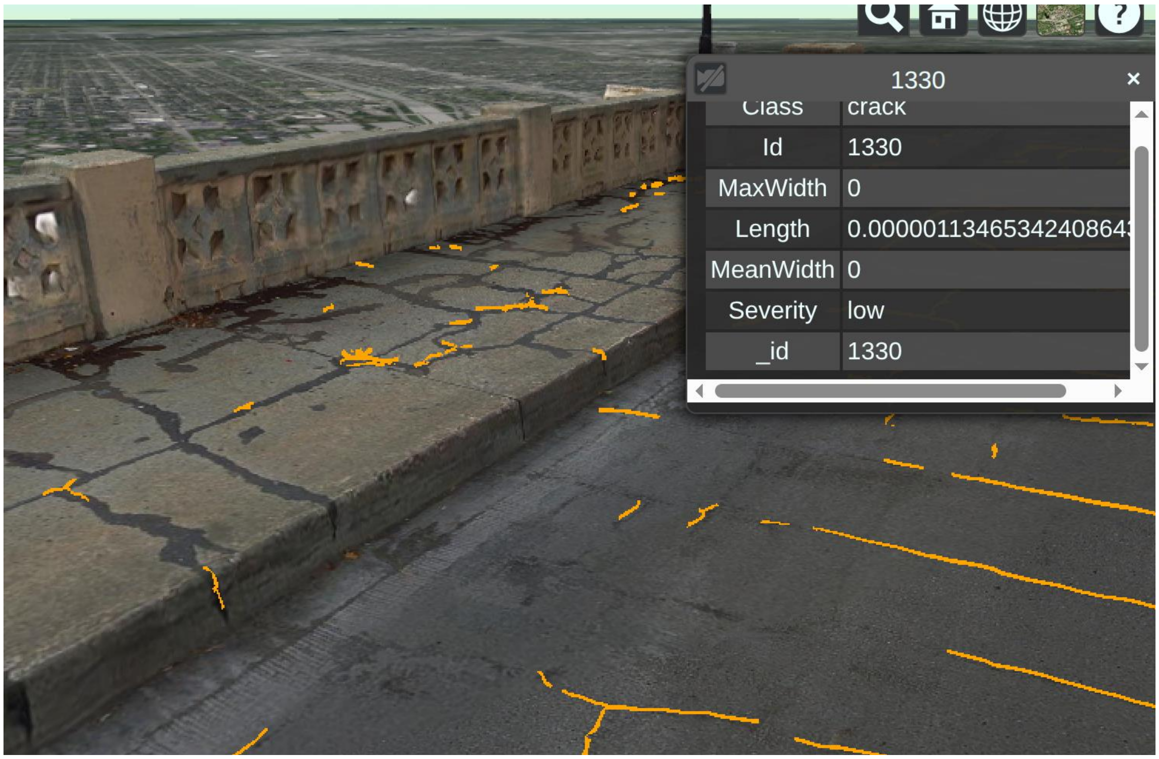
- https://www.swisstopo.admin.ch/en/swissbedrock-en
- https://docs.geo.admin.ch/visualize-data/3d-tiles.html

Is implementation complete? ☒ Yes ☐ No

3. Implementation name: CesiumJS

Date of most recent version: Ongoing as the 3D Tiles 2.0 spec finalizes

Implementation description: Open-source reference implementation with support for all 3D Tiles 2.0 capabilities. Experimental features are expected to stabilize.



Vector polylines using 3D Tiles 2.0 and rendered in CesiumJS. Source data: Collins Engineers, Inc.

Implementation URL:

- https://github.com/CesiumGS/cesium
- https://sandcastle.cesium.com/?id=voxels-in-3d-tiles
- https://sandcastle.cesium.com/?id=3d-tiles-gaussian-splats-with-lod

Is implementation complete? ☐ Yes ☒ No

If not, what portions of the proposed Community standard are implemented?

All features except glTF 2.1 are fully implemented. The glTF 2.1 implementation is partially complete as the glTF 2.1 specification evolves and will be fully completed by full ratification of 3D Tiles 2.0.

4. Implementation name: Cesium Native / Cesium for Unreal

Date of most recent version: Ongoing as the 3D Tiles 2.0 spec finalizes

Implementation description: Open-source C++ streaming library and serializers in Cesium Native. Open-source renderers in Cesium for Unreal and Cesium for Unity. Experimental features are expected to stabilize.

3D Gaussian splats rendered in Cesium for Unreal.

Implementation URL:

- https://github.com/CesiumGS/cesium-native
- https://github.com/CesiumGS/cesium-unreal

Is implementation complete? ☐ Yes ☒ No

If not, what portions of the proposed Community standard are implemented?

Voxels, Vector Data, and 3D Gaussian splatting

5. Implementation name: Cesium Tiling Pipelines

Date of most recent version: Ongoing as the 3D Tiles 2.0 spec finalizes

Implementation description: Closed-source tiling pipelines convert digital elevation models, massive photogrammetry models, point clouds, and CityGML to 3D Tiles 2.0. Open-source 3D Tiles 2.0 samples in progress.

High-level diagram showing how design data is merged with geospatial content through 3D Tiles and the Cesium Tiling Pipeline.

Implementation URL:

- https://github.com/CesiumGS/3d-tiles-samples

Is implementation complete? ☐ Yes ☒ No

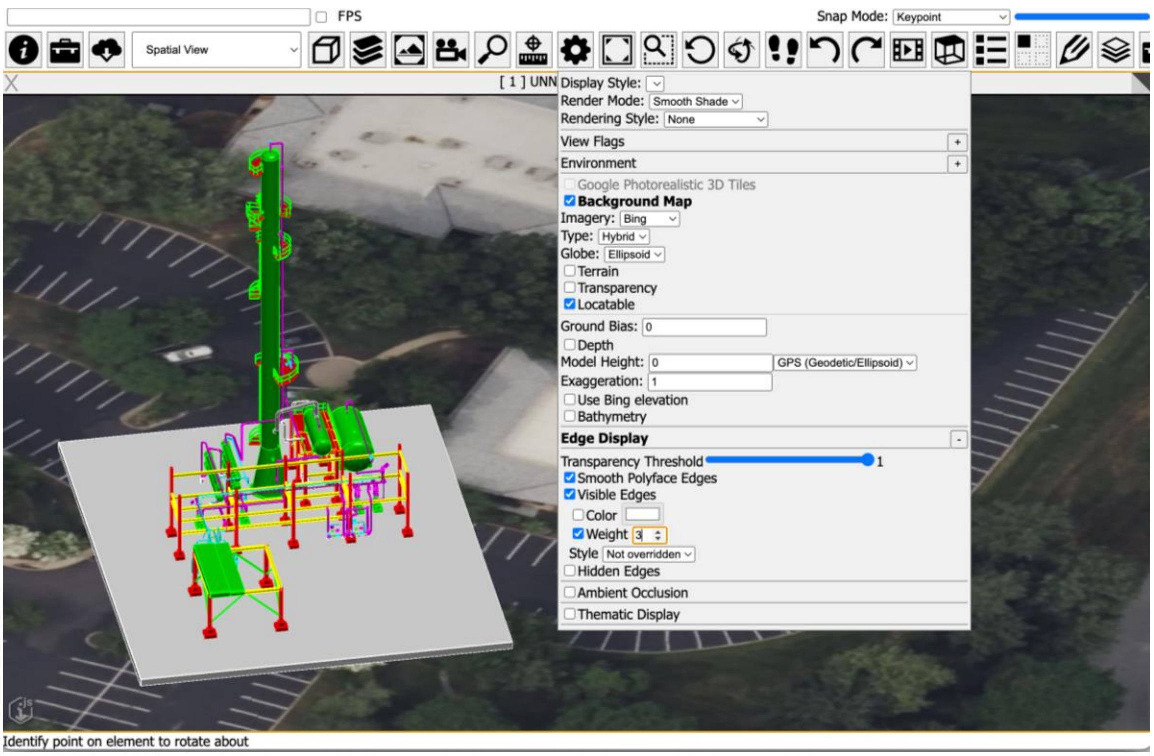
If not, what portions of the proposed Community standard are implemented?

Tiling Vector data and 3D Gaussian splatting data support is fully implemented. The glTF 2.1 implementation is partially complete as the glTF 2.1 specification fully evolves and will be fully completed by full ratification of 3D Tiles 2.0.

6. Implementation name: iTwin.js

Date of most recent version: Ongoing as the 3D Tiles 2.0 spec finalizes

Implementation description: Visualization library that combines 3D Tiles 2.0 with iModel aggregated data streams for rendering digital twins.



A glTF (.glb) file with the EXT_mesh_primitive_edge_visibility extension, loaded into iTwin.js and rendering the encoded edges alongside the surface geometry.

Implementation URL:

- https://github.com/iTwin/itwinjs-core

Is implementation complete? ☐ Yes ☒ No

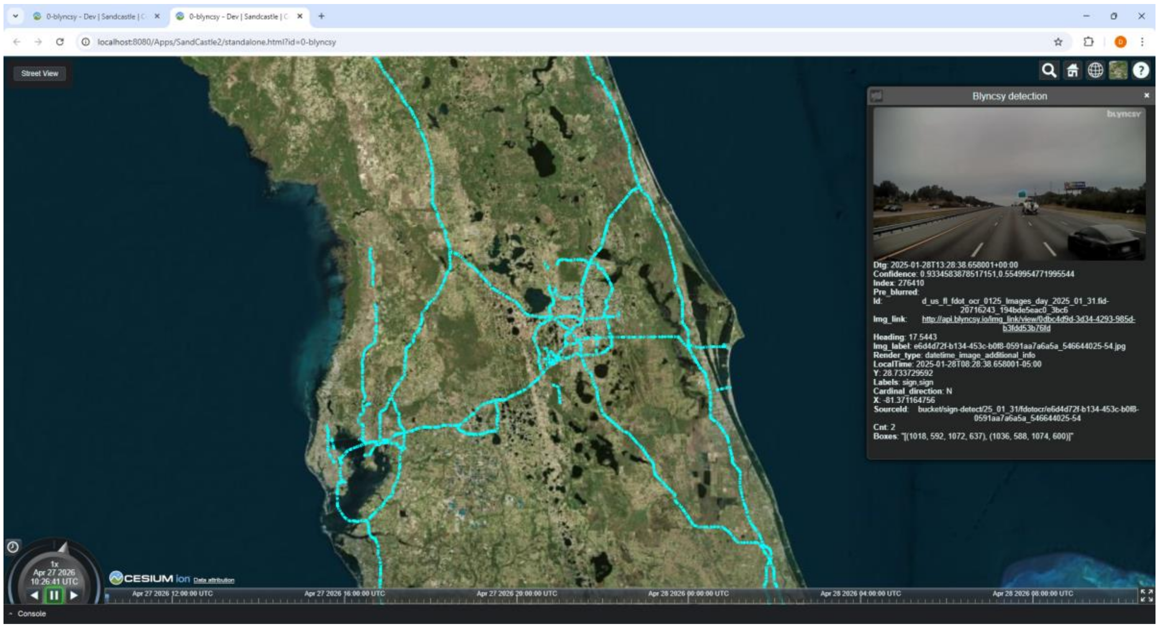
If not, what portions of the proposed Community standard are implemented?

AEC rendering extensions

7. Implementation name: Blyncsy

Date of most recent version: Ongoing as the 3D Tiles 2.0 spec finalizes

Implementation description: Blyncsy uses machine learning to help departments of transportation automate roadway maintenance and asset inventory activities. Inspections and issues are mapped using vector map data using machine-learning powered dashcam analysis.



Blyncsy Data in CesiumJS Sandcastle

Implementation URL:

- https://blyncsy.com/

Is implementation complete? ☐ Yes ☒ No

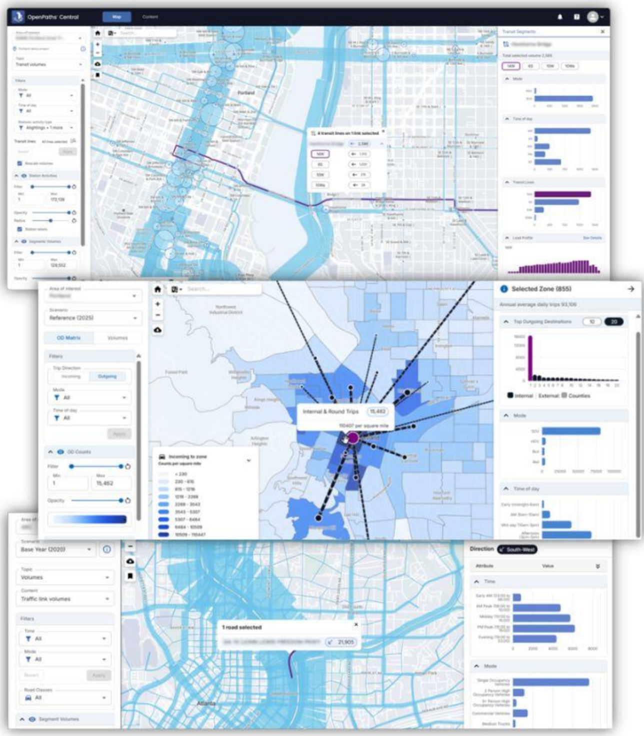
If not, what portions of the proposed Community standard are implemented?

Vector Tiles

8. Implementation name: OpenPaths

Date of most recent version: Ongoing as the 3D Tiles 2.0 spec finalizes

Implementation description: OpenPaths is a comprehensive transport modeling platform for strategic and operational transport planning. Using vector tiles, it enables large geo-spatially oriented data sets of multi-modal transport network modeling, travel demand forecasting, and scenario planning.



OpenPaths Central

Implementation URL:

- https://central.openpaths.bentley.com/

Is implementation complete? ☐ Yes ☒ No

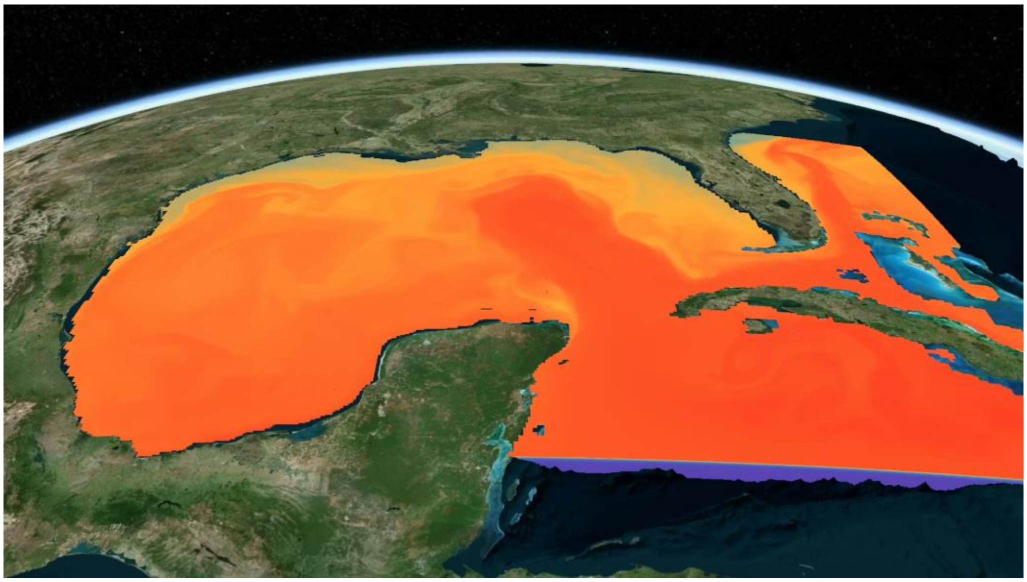
If not, what portions of the proposed Community standard are implemented?

Vector Tiles

9. Implementation name: UTIC

Date of most recent version: Ongoing as the 3D Tiles 2.0 spec finalizes

Implementation description: Uses voxels in 3D Tiles 2.0 to stream large underwater and maritime datasets such as ocean temperature to simulate Naval combat scenarios directly in the browser.



Water temperatures in the Gulf of Mexico represented using 3D Tiles 2.0 Voxels and CesiumJS.

Implementation URL:

- https://www.underseatech.org/

Is implementation complete? ☐ Yes ☒ No

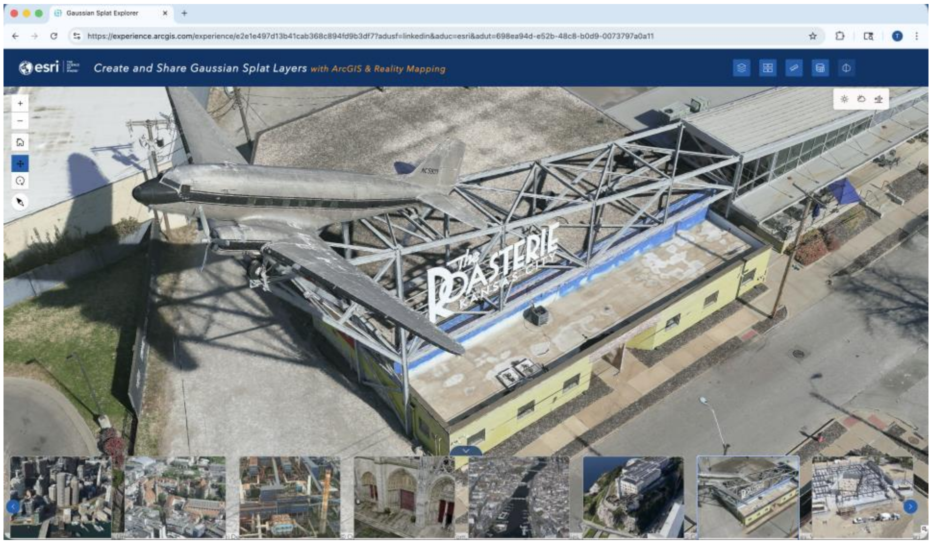
If not, what portions of the proposed Community standard are implemented?

Voxels

10. Implementation name: Esri

Date of most recent version: Ongoing as the 3D Tiles 2.0 spec finalizes

Implementation description: Esri's ArcGIS Platform implements 3D Tiles across creation, hosting, and consumption workflows in web, desktop, and mobile environments — including ArcGIS Pro 3.6 and ArcGIS Online — with support for the 3D Tiles 2.0 extension for 3D Gaussian Splatting.



3D Gaussian splats in Esri using 3D Tiles.

Implementation URL:

- https://www.esri.com/en-us/home
- https://experience.arcgis.com/experience/e2e1e497d13b41cab368c894fd9b3df7?adusf=linkedin&aduc=esri&adut=698ea94d-e52b-48c8-b0d9-0073797a0a11

Is implementation complete? ☐ Yes ☒ No

If not, what portions of the proposed Community standard are implemented?

3D Gaussian splatting

11. Implementation name: EARTHBRAIN

Date of most recent version: Ongoing as the 3D Tiles 2.0 spec finalizes

Implementation description: EARTHBRAIN uses time-dynamic 3D Tiles to visualize how earthworks construction sites change over time. A tileset is incrementally updated as new drone survey and as-built data is collected and is played back in CesiumJS. Support for this is being added to the next version of Smart Construction Dashboard.

Time-dynamic 3D Tiles with Smart Construction data.

Implementation URL:

- https://smartconstruction.io/

Is implementation complete? ☐ Yes ☒ No

If not, what portions of the proposed Community standard are implemented?

Time-dynamic 3D Tiles

12. Implementation name: Hexagon

Date of most recent version: Ongoing as the 3D Tiles 2.0 spec finalizes

Implementation description: Hexagon visualizes a myriad of 3D data across many Geospatial industries including Aerospace, Mining, Oil & Gas, and many more. Hexagon supports 3D Tiles 2.0 for 3D Gaussian splatting, advanced metadata, and other use cases.

Implementation URL:

- https://hexagon.com/

Is implementation complete? ☒ Yes ☐ No

6. Public availability

Is the proposed Community standard currently publicly available? ☒ Yes ☐ No

URL: https://github.com/CesiumGS/3d-tiles/tree/main/next/2.0

7. Supporting member(s)

- Ansys
- Bentley Systems, Inc.
- Camptocamp
- Ecere
- Epic Games, Inc.
- Hexagon
- Octave
- Pacific Spatial Solutions
- SwissTopo
- Trimble
- UK Ordnance Survey
- U.S. Army Geospatial Center (AGC)
- U.S. National Air and Space Agency (NASA)
- U.S. National Geospatial-Intelligence Agency (NGA)

8. Intellectual property rights

Will the contributor retain intellectual property rights? ☒ Yes ☐ No

If yes, the contributor will be required to work with OGC staff to properly attribute the submitter’s intellectual property rights.

If no, the contributor will assign intellectual property rights to the OGC.